

KUSHAGRA JAISWAL

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Education

MIT World Peace University <i>B.Tech in Computer Science and Engineering</i> <i>CGPA: 7.47 / 10</i>	Pune, India <i>July 2023 – June 2027</i>
Guru Nanak Public School, Dalhousie <i>XII CBSE: 85.2%</i>	<i>2021 – 2022</i>
Guru Nanak Public School, Dalhousie <i>X CBSE: 97%</i>	<i>2019 – 2020</i>

Projects

BarPal.ai - Legal RAG LLM Assistant <i>Python, FastAPI, Pinecone, Gemma 4 (31B)</i> GitHub	Nov 2025
- Built a Generative AI powered legal assistant using semantic RAG architecture with Pinecone Vector Database, NVIDIA Inference API, and Gemma 4 31B for Q&A over PDFs and DOCX files. Slashed RAM usage by 90% (from 800MB to under 80MB) and eliminated Out-of-Memory (OOM) by switching from PyTorch local embedding to NVIDIA Inference API - Accelerated document ingestion speed, achieving a 60%+ reduction in chunk-embedding latency by architecting a parallel-batched ingestion framework using asyncio.gather - Built an LLM Evaluation pipeline using 3 RAGAS metrics (Faithfulness, Answer Relevance, Context Precision)	
NAT Funds - Mutual Fund Analytics Platform <i>Node.js, Express.js, Tailwind CSS, REST APIs</i> GitHub	Apr 2026
- Built a full-stack mutual fund analytics platform to discover, filter, and compare 9,100+ Indian mutual funds by aggregating live data from AMFI, NSE/BSE Indices, and RBI APIs. - Developed 15+ financial metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta and Standard Deviation using real benchmarks. Reduced cold-boot latency by 80% on warm restarts via a 3-tier disk-backed caching system with 4 scheduled cron jobs for automated daily market data updates.	
Solar Panel Fault Detection <i>Python, TensorFlow, Keras, Scikit-learn</i> GitHub	Apr 2026
- Built a ML model to classify 12 fault categories in solar panels using CNN and Random Forest on a 20,000-image RaptorMaps dataset. - Achieved 75% test accuracy and 90% Top-2 accuracy on a held-out test set by applying a 2-phase fine-tuning strategy. - Applied Batch Normalisation, L2 regularisation, and Dropout to reduce overfitting - Integrated Grad-CAM visualisations to highlight fault regions.	
LocalDrop - LAN File Sharing System <i>Java, Servlets, JSP, HikariCP, MySQL</i> GitHub	Mar 2026
- Built a browser-based LAN file-sharing server enabling devices on the same Wi-Fi to upload, download, search, and track transfers without client software. - Handled file transfers up to 500 MB via 8 KB buffered streaming with SHA-256 integrity verification. Secured against SQL injection using PreparedStatements - Optimised for 10 concurrent connections with a HikariCP connection pool.	

Technical Skills

Languages	Python, Java, JavaScript, SQL
CS Fundamentals	Data Structures, OOP, DBMS, Computer Networks, Operating Systems, Software Engineering, Agile
Backend & APIs	AI APIs, FastAPI, REST APIs, Node.js, Express.js, MySQL, PostgreSQL (Basic)
AI/ML	Generative AI, Agentic AI, LLMs, NLP, RAG Pipelines, Vector Database, Transformers, Deep Learning, Prompt Engineering, LLM Evaluation, Machine Learning
Libraries	Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Langchain(Basic), React.js
Tools	Git, GitHub, Hugging Face, AWS (EC2, S3 basics), Docker, Tableau
Certifications	IBM Full Stack Web Development, Google 5-Day AI Agents, Kaggle Intermediate Machine Learning, JPMorganChase - Software Engineering

Extracurricular Activities

Videographer & Video Editor MIT-WPU Shutterbugs (Media Club)	2024 – Present
Media Creator MIT-WPU SquadUP (Esports Club) Created media and covered government-college collaboration events such as Maratha Esports.	2024 – Present
Outreach Leader Innovation Hub Served as Outreach Leader for RTEP, a national-level research paper conference.	2023